AdifySkills

n8n Automation Mastery
+91 83838 91716

CLASS NOTES

Module 1 · Automation Foundations

DAY 03 of 25

n8n Data Model: Items, JSON & Binary

"Jo data samajh gaya, woh n8n samajh gaya."

⚡ AAJ HUM YEH BANAYENGE

CSV Sales → City-wise Report Email

Form par sales ki CSV file upload karo. Workflow file padhta hai, city-wise total nikaalta hai aur ek saaf HTML table email karta hai.

Split Out | Aggregate | Summarize | Extract from File

Naam: _____ Class date: _____

2 ghante ki live class · 7:00 PM – 9:00 PM IST

🧭 Aaj ka roadmap

Ek nazar mein poori class — kya, kyun aur kitne time mein.

💡 Yeh lesson kyun zaroori hai?

n8n mein sab kuch "items" hai. Jo student items ko samajh gaya, usse koi bhi workflow mushkil nahi lagega.

🎯 Aaj ke baad aap yeh kar paoge

01 JSON basics: objects, arrays, nesting

🌍 Visiting card: Naam, Phone, Email — har field ka ek label aur value.
⚙️ {"name": "Aman", "phones": ["98...", "99..."], "address": {"city": "Pune"}}

02 n8n items — har node items ki list process karta hai

🌍 Conveyor belt par 10 boxes — machine har box par same kaam karti hai.
⚙️ HTTP se 10 users aaye → Edit Fields har user par chala → 10 output items.

03 Item linking & paired items

🌍 Order slip aur uska bill — dono ek doosre se jude hain.
⚙️ Paired items se pata rehta hai ki output item kis input item se bana.

04 Binary data: files, images, PDFs

🌍 Courier mein letter (JSON) + parcel attached (binary).
⚙️ Gmail attachment → binary.data → Extract from File → text.

05 Split Out, Aggregate & Summarize nodes

🌍 Mithai ka dabba kholna (split), wapas bharna (aggregate), ginti karna (summarize).
⚙️ Order ke line items Split Out → Summarize by product → Aggregate into one email.

06 Data pinning & mock data for testing

🌍 Rough copy par practice — final copy kharab nahi hoti.
⚙️ API ka output pin karo → aage ke nodes bina API call ke test karo.

⏱ Class ka plan (2 ghante)

10m	40m	40m	20m	10m

0:00-0:10 Recap

🪄 Cricket match se pehle warm-up karte ho na? Recap dimaag ka warm-up hai.

0:10-0:50 Concept: Data Structure

Visual walkthrough of items flow

🪄 Teacher ke paas 30 copies hain. Woh har copy par same kaam karti hai — check karna, marks dena. n8n bhi har item par wahi karta hai.

0:50-1:30 Live Build

CSV upload → clean → split → aggregate report

🪄 Aaj ki kahani yaad hai — 'Tiffin factory ki assembly line'? Ab wahi kahani n8n mein banayenge. Har scene ek node hai: ek-ek block jodo aur har block ke baad dekho sab chal raha hai ya nahi.

1:30-1:50 Quiz + Challenge

Module 1 quiz (15 questions)

🪄 Practice match — jitna yahan girenge, utna asli match mein kam girenge.

1:50-2:00 Q&A + Homework

🪄 Ghar jaane se pehle bag check — sab samajh aaya? Homework diary mein likha?

Concepts — aasaan bhasha mein

🧠 Har concept ke saath ek real-life analogy. Analogy yaad rahegi to concept bhi yaad rahega.

CONCEPT 01

JSON

Data likhne ka format: { "key": "value" }. Objects {} aur arrays [].

🪄 **Analogy:** Form ke fields aur unke jawab

CONCEPT 02

Items

Har node items ki LIST leta hai aur har item par kaam karta hai.

🪄 **Analogy:** Assembly line par boxes

CONCEPT 03

Split Out vs Aggregate

Split Out: 1 item ki list → kai items. Aggregate: kai items → 1 list.

🪄 **Analogy:** Packet kholna vs packet bharna

CONCEPT 04

Binary data

Files (PDF, image) JSON ke saath "binary" property mein chalti hain.

🪄 **Analogy:** Courier ke saath attached parcel

🔑 Aaj ka main idea

n8n mein data "items" ki list ke roop mein chalta hai. Har item ek JSON object hai — jaise ek form ka ek jawab. Agar 10 items aaye, to agla node 10 baar chalega. Yeh samajh aa gaya to n8n 50% aa gaya.

- Attendance register: har line = ek student = ek item
- Shopping list: har cheez = ek item
- WhatsApp group ke 20 messages = 20 items

📖 Ek kahani se samjho: Tiffin factory ki assembly line

1. 📦 **Ek bada dabba aaya (CSV)** = Extract from File
2. 📤 **Dabba khola, 50 tiffin nikle** = Split Out → 50 items
3. 🧹 **Toote tiffin hataye** = Filter
4. 🏙️ **City-wise ginti** = Summarize
5. 🚚 **Report bhej di** = Gmail

🪄 Ek line mein: “Teacher ke paas 30 copies hain. Woh har copy par same kaam karti hai — check karna, marks dena. n8n bhi har item par wahi karta hai.”

🗺️ Aaj ka workflow: CSV Sales → City-wise Report Email

Data left se right chalta hai. Har dabba ek n8n node hai.



💰 Isse kitna kama sakte ho?

Distributors, D2C brands (daily/weekly sales report) — aam taur par **₹5,000 – ₹12,000**. *(Andaaza hai; client aur scope ke hisaab se badalta hai.)*

🛠 Step-by-step build — n8n kholo aur saath chalo

1. ⚡ **Manual Trigger**
   Testing ke liye
2. 🌐 **HTTP Request**
   CSV file URL se download (Response: File)
3. 📄 **Extract from File**
   Operation: Extract from CSV → har row ek item
4. 🔀 **Filter**
   amount > 0 (khaali/galat rows hatao)
5. 📈 **Summarize**
   Group by: city, Sum: amount, Count: orders
6. 📧 **Gmail**
   City-wise summary table email

✅ Har step ke baad **Execute step** dabao aur output check karo. Sab green ho to workflow **Active** karo.

🔑 Setup (import ke baad)

- ☐ Connect: **Gmail (Google OAuth)**
- ☐ Sample CSV download karo (neeche link) — columns: order_id, city, amount

✅ Test kaise karein

1. Form ka Test URL kholo → sample-sales.csv upload karo
2. Har node ka output dekho: items kaise group hue
3. Email mein table check karo

⚖️ Pehle vs Ab — farak dekho

Yahi farak client ko dikhana hai. Time bachta hai = paisa bachta hai.

😩 Pehle (manual)

Excel kholna	2 min
City-wise filter + total	20 min
Summary type karna	10 min
Email bhejna	3 min
	35 min

➜

⚡ Ab (n8n)

CSV aate hi summary email

0 min

⏱ Har report par 35 min bache, calculation galti zero.

⌨️ Aaj ka key snippet

Copy karke n8n mein paste karo 📋 Select karke copy karo

```
{{ $input.all().length }}  // kitne items aaye
```

📦 Sample data — input se output tak

📥 Input 📋 Select karke copy karo

```
[
  {
    "city": "Delhi",
    "amount": 1200
  },
  {
    "city": "Pune",
    "amount": 800
  },
  {
    "city": "Delhi",
    "amount": 500
  }
]
```

📤 Output 📋 Select karke copy karo

```
[
  {
    "city": "Delhi",
    "sum_amount": 1700,
    "count": 2
  },
  {
    "city": "Pune",
    "sum_amount": 800,
    "count": 1
  }
]
```

💬 Copy-paste prompts

ChatGPT / Claude mein seedha paste karo. PDF mein text select karke copy ho jata hai.

💬 Build it with ChatGPT / Claude 📋 Select karke copy karo

```
Act as an n8n expert and teacher. Help me build this workflow in n8n step by step (n8n 1.x UI):

Goal: Form par sales ki CSV file upload karo. Workflow file padhta hai, city-wise total nikaalta hai aur ek saaf HTML table email karta hai.

Flow: Form (file upload) (sales.csv) → Extract from File (CSV → items) → Code (City-wise sum) → Code (HTML table) → Gmail (Report email)

For each node tell me: which node to add, exact settings, and the expressions to use. Then tell me how to test it and the 3 most common errors beginners hit.
```

💬 Debug helper prompt 📋 Select karke copy karo

```
I'm building the n8n workflow "CSV Sales → City-wise Report Email" (nodes: Upload Sales CSV, Read CSV, City-wise Total, Build HTML Table, Email Report).
Node "<NODE NAME>" gives this error:
<PASTE ERROR>

Input JSON of that node:
<PASTE INPUT>

Explain the cause in simple words and give me the exact fix (settings or expression).
```

📥 Workflow ko n8n mein kaise laayein

1. **Workflow JSON** wale button par click karo (ya lesson page par 📋 **Copy workflow JSON** dabao).
2. Poora text copy karo (`Ctrl` + `A` , `Ctrl` + `C`).
3. n8n mein naya workflow kholo → canvas par click → `Ctrl` + `V` . Ya … menu → *Import from File*.
4. Upar wala Setup checklist poora karo aur **Test** dabao.

⚡ Workflow JSON
5 nodes · ready to import
`adifyskills.com/workflows/day-03-csv-city-sales-report.json`

🌐 Lesson page
Copy buttons, quiz aur video links
`adifyskills.com/day/3`

🧯 Atak gaye? Yeh check karo

Error aana seekhne ka hissa hai — sab atakte hain. Neeche wale points se 90% problems solve ho jaati hain.

Agar yeh ho raha hai…	To yeh karo
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "Gmail (Google OAuth)" connect karo.
Webhook/Form test par kuch nahi ho raha?	"Listen for test event" / Test URL use karo aur 2 minute ke andar request bhejo. Production URL sirf Active workflow par chalta hai.

⚠️ Yeh galtiyan sabse common hain

- ! Ek item expect karna jab 50 items aa rahe hon (har email 50 baar jayega!)
- ! Nested array ko bina Split Out ke process karna
- ! Binary property ka naam galat dena (data vs file)

⭐ Pro tips — time bachane wali tricks

- ★ Output panel mein Table / JSON / Schema views badal ke dekho
- ★ Item count node ke upar dikhta hai — hamesha check karo
- ★ Aggregate se pehle Sort karo agar order matter karta hai

Phir bhi atke? Koi baat nahi 🙂
Screenshot + error message ke saath WhatsApp par poochho →

🏋️ Practice + Challenge

Padhne se zyada banane se seekhte hain. Ek-ek level upar jaao.

🛠 Live lab

CSV sales file ko process karke city-wise summary nikaalna aur email karna.

📝 Homework

Public API (e.g. JSONPlaceholder) se data laake usse Split Out + Aggregate se transform karo.

- ☐ Homework ho gaya

BEGINNER
Sirf Delhi ke orders filter karo

INTERMEDIATE
Top 3 cities sort + limit

ADVANCED
Summary ko Excel file bana ke attach karo

🧩 Aaj ka challenge — pehle khud try karo

🎯 Report mein sirf woh cities dikhao jinki sales ₹5,000 se zyada hai.

💡 **Hint 1**
Code node ke baad ek naya node lagao jo items ko chhaant de.

💡 **Hint 2**
Filter node mein condition: sales (number) "is greater than" 5000.

✅ Solution aakhri page par hai — pehle 5 minute khud socho 😉

🎯 2 minute ka self-check

Sahi option par gol ghera lagao. Answers aakhri page par — score sirf aapke liye hai.

Q1. 10 items ek Gmail node mein jaayein to kitne emails?

(A) Error
(B) 1
(C) 10
(D) 0

Q2. Kai items ko ek list banane ka node?

(A) Filter
(B) IF
(C) Split Out
(D) Aggregate

Q3. PDF file n8n mein kahan hoti hai?

(A) binary
(B) headers
(C) credentials
(D) json

Q4. JSON mein list kis bracket mein?

(A) {}
(B) []
(C) ()
(D) <>

📖 Aaj ke naye shabd

Workflow
Steps ki ek chain jo apne aap chalti hai — jaise recipe ke steps.

Node
Workflow ka ek dabba/step. Har node ek kaam karta hai (email bhejna, data laana…).

Trigger
Jo workflow ko START karta hai — jaise alarm, naya email, ya form submit.

JSON
Data likhne ka format: { "naam": "Ravi", "city": "Delhi" } — key aur value ke jode.

Item
Data ki ek row/record. n8n mein data items ki list ki tarah flow karta hai.

Binary
File wala data — PDF, image, CSV. Text se alag hota hai.

HTTP Request
Node jo kisi bhi API ko call kar sakta hai — n8n ka universal connector.

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+91 83838 91716

✍️ Meri notes

Class mein jo naya laga, jo doubt aaya — yahan likho. Kal revise karna aasaan hoga.

⏭ Kal kya seekhenge — Day 4

Expressions & Data Transformation

"{{ }} — n8n ki asli superpower."

☐ **Day 3 complete! 🎉**

Tick karo aur website par bhi ✅ dabao.

🔑 Answer key + useful links

Pehle khud try kiya? Ab check karo. Galat hua to explanation padho — wahi asli seekh hai.

Q1 → C
10
Har item ke liye node ek baar chalta hai.

Q2 → D
Aggregate

Q3 → A
binary

Q4 → B
[]

✅ Challenge ka solution

City-wise Total ke baad Filter node: {{ $json.sales }} → Number → is greater than → 5000. Phir Build HTML Table.

📚 Official docs

- Data structure ↗
- Binary data ↗
- Data pinning ↗
- Data mapping ↗
- Item linking ↗

▶ Videos

- n8n items data structure explained ↗
- n8n split out aggregate ↗

💬 Doubt poochho
WhatsApp: +91 83838 91716
wa.me/918383891716

🌐 Day 3 lesson page
Quiz, copy buttons, videos
adifyskills.com/day/3

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